Linux Quick Reference

College of Technology

ITT-221

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#10, MedS1, MedW1

Server: 172.10.0.1

Desktop: 172.10.100.1

Command	**Outcome**

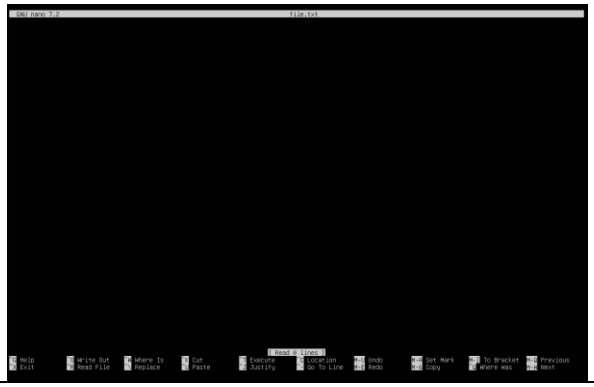
Pwd	Current Location
	`linuxuser@linuxserver:/$ pwd`
	`/`
	`linuxuser@linuxserver:/$`

Ls	Equals directory, shows contents of current directory
	`linuxuser@linuxserver:/$ ls`
	`bin boot dev home lib64 lost+found mnt proc run sbin.usr-is-merged srv sys usr`
	`bin.usr-is-merged cdrom etc lib lib.usr-is-merged media opt root sbin snap swap.img tmp var`
	`linuxuser@linuxserver:/$`

Cd	Change directory
	`linuxuser@linuxserver:/$ cd /home`
	`linuxuser@linuxserver:/home$ _`

Sudo	Supervisor Do (what's next)
	`linuxuser@linuxserver:~$ sudo mv file.txt /home`
	`[sudo] password for linuxuser:`
	`linuxuser@linuxserver:~$ _`

Touch <filename>	Create file, update timestamp
	`linuxuser@linuxserver:~$ touch file.txt`
	`linuxuser@linuxserver:~$`

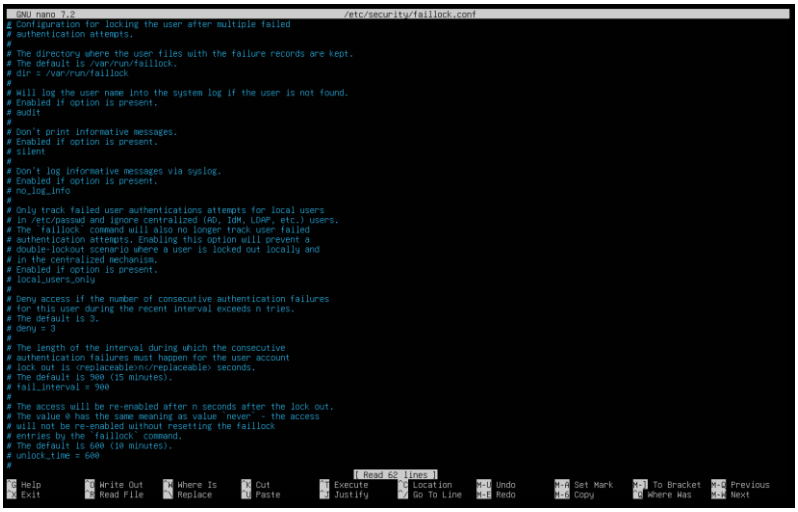
Nano <filename>	Create file & opens it in nano Example: Nano file.txt 
Vi <filename>	Ditto – opens in Vi Example: Vi file.txt
Mv <filename> <Destination Directory>	Move file to… linuxuser@linuxserver:~$ sudo mv file.txt /home [sudo] password for linuxuser: linuxuser@linuxserver:~$ _

Cp <filename> <Destination> <New Filename>	Copy file to… ```linuxuser@linuxserver:~$ sudo cp /home/file.txt /bin linuxuser@linuxserver:~$ _``` (screenshot)
History	Lists recent commands ```163 /home/linuxuser/full_backup.sh 164 chmod +x /home/linuxuser/full_backup.sh 165 chmod +x /home/linuxuser/incremental_backup.sh 166 /home/linuxuser/full_backup.sh 167 ls /var/backups 168 /home/linuxuser/full_backup.sh 169 crontab -l 170 systemctl status cron 171 nano full_backup.sh 172 crontab -e 173 /home/linuxuser/full_backup.sh 174 mkdir -p /home/linuxuser/backups 175 nano /home/linuxuser/full_backup.sh 176 /home/linuxuser/full_backup.sh 177 ls -l /home/linuxuser/backups 178 nano /home/linuxuser/incremental_backup.sh 179 chmod +x /home/linuxuser/incremental_backup.sh 180 /home/linuxuser/incremental_backup.sh 181 ls -l /home/linuxuser/backups 182 crontab -e 183 crontab -l 184 cat /etc/crontab 185 ls /etc/cron.daily 186 ls /etc/cron.weekly 187 cat /etc/crontab 188 cat /etc/cron.daily/* 189 cat /etc/cron.weekly/* 190 cat /etc/crontab 191 cat /etc/cron.daily/* 192 cat /etc/cron.weekly/* 193 crontab -l 194 nano /home/linuxuser/incremental_backup.sh 195 crontab -l 196 cat /etc/crontab 197 cat /etc/cron.daily/* 198 cat /etc/cron.weekly/* 199 nano /home/linuxuser/incremental_backup.sh 200 nano /home/linuxuser/full_backup.sh 201 logout 202 touch file.txt 203 nano file.txt 204 vi file.txt 205 mv file.txt /home 206 sudo mv file.txt/home 207 sudo mv file.txt /home 208 sudo cp file.txt /tmp 209 sudo cp file.txt /bin 210 sudo cp /home/file.txt /bin 211 history linuxuser@linuxserver:~$ _``` (screenshot)
Ls-l	Show contents; additional info; permissions; owner ```linuxuser@linuxserver:~$ ls -l total 12 drwxrwxr-x 2 linuxuser linuxuser 4096 Apr 4 20:02 backups -rwxrwxr-x 1 linuxuser linuxuser 93 Apr 4 20:00 full_backup.sh -rwxrwxr-x 1 linuxuser linuxuser 288 Apr 4 20:13 incremental_backup.sh linuxuser@linuxserver:~$ _``` (screenshot)
Ip addr show	Ipconfig/all ```linuxuser@linuxserver:~$ ip addr show 1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000 link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 inet 127.0.0.1/8 scope host lo valid_lft forever preferred_lft forever inet6 ::1/128 scope host noprefixroute valid_lft forever preferred_lft forever 2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000 link/ether 00:0c:29:e3:f1:f8 brd ff:ff:ff:ff:ff:ff altname enp2s1 inet 192.168.196.129/24 metric 100 brd 192.168.196.255 scope global dynamic ens33 valid_lft 1226sec preferred_lft 1226sec inet6 fe80::20c:29ff:fee3:f1f8/64 scope link valid_lft forever preferred_lft forever 3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000 link/ether 00:0c:29:e3:f1:02 brd ff:ff:ff:ff:ff:ff altname enp2s5 inet 172.10.0.1/16 brd 172.10.255.255 scope global ens37 valid_lft forever preferred_lft forever inet6 fe80::20c:29ff:fee3:f102/64 scope link valid_lft forever preferred_lft forever linuxuser@linuxserver:~$ _``` (screenshot)

Ip-br address	Ipconfig (easier to read) ```linuxuser@linuxserver:~$ ip -br address lo UNKNOWN 127.0.0.1/8 ::1/128 ens33 UP 192.168.196.129/24 metric 100 fe80::20c:29ff:fee3:f1f8/64 ens37 UP 172.10.0.1/16 fe80::20c:29ff:fee3:f102/64 linuxuser@linuxserver:~$``` Ip route Show the routing table ```linuxuser@linuxserver:~$ ip route default via 192.168.196.2 dev ens33 proto dhcp src 192.168.196.129 metric 100 172.10.0.0/16 dev ens37 proto kernel scope link src 172.10.0.1 192.168.196.0/24 dev ens33 proto kernel scope link src 192.168.196.129 metric 100 192.168.196.2 dev ens33 proto dhcp scope link src 192.168.196.129 metric 100 linuxuser@linuxserver:~$ _``` Sudo ip address add n.n.n.n/n dev <portname> Config IP address Example: Adding the IP address I already have ```linuxuser@linuxserver:~$ sudo ip address add 172.10.0.1/16 ens37 Error: either "local" is duplicate, or "ens37" is a garbage. linuxuser@linuxserver:~$``` Adduser <username> Creates a high-level user account - Creates a profile - Creates a home directory ```linuxuser@linuxserver:~$ sudo adduser example info: Adding user `example' ... info: Selecting UID/GID from range 1000 to 59999 ... info: Adding new group `example' (1003) ... info: Adding new user `example' (1003) with group `example (1003)' ... info: Creating home directory `/home/example' ... info: Copying files from `/etc/skel' ... New password: Retype new password: passwd: password updated successfully Changing the user information for example Enter the new value, or press ENTER for the default Full Name []: example Room Number []: example Work Phone []: example Home Phone []: example Other []: example Is the information correct? [Y/n] y info: Adding new user `example' to supplemental / extra groups `users' ... info: Adding user `example' to group `users' ... linuxuser@linuxserver:~$ _```

Useradd	Creates a low-level user account - Does not set a password - Does not create a directory webdevs:x:1003: linuxuser@linuxserver:~$ sudo useradd user1 linuxuser@linuxserver:~$ sudo useradd user2 linuxuser@linuxserver:~$ sudo useradd user3 linuxuser@linuxserver:~$ _
Sudo usermod -aG sudo <username>	Makes the account a SUDO account linuxuser@linuxserver:~$ sudo usermod -aG webdevs user1 linuxuser@linuxserver:~$ sudo usermod -aG webdevs user2 linuxuser@linuxserver:~$ sudo usermod -aG webdevs user3 linuxuser@linuxserver:~$
Sudo-u <username><command>	Allows sudo changes in a lower account/other account linuxuser@linuxserver:~$ sudo -u nobody whoami nobody linuxuser@linuxserver:~$
Sudo getent group sudo	Get members of the SUDO group linuxuser@linuxserver:~$ sudo getent group sudo sudo:x:27:linuxuser,otto.server linuxuser@linuxserver:~$ _
Getent group <groupName>	Get members of a group linuxuser@linuxserver:~$ sudo getent group users users:x:100:otto.server,bubblefrog,example linuxuser@linuxserver:~$
Chown <owner> <filename>	Change ownership (owner trumps group) linuxuser@linuxserver:/var/www/html$ sudo chown root:webdevs *

Chown <owner>:<group> <filename>	Change ownership and group (owner trumps group) ```linuxuser@linuxserver:~$ sudo chown nobody:nogroup /usr/local/bin/testfile.zip linuxuser@linuxserver:~$ ls -l /usr/local/bin/testfile.zip -rw-r--r-- 1 nobody nogroup 118498 Apr 12 21:06 /usr/local/bin/testfile.zip linuxuser@linuxserver:~$ _```
Chmod <permissions> <filename>	Change permissions on a file (read = 4, write = 2, execute = 1) Example: chmod 764 = d rwx rw- r— Owner-Group-All Users ```linuxuser@linuxserver:/var/www/html$ sudo chmod 644 *```
Chmod u + x filename.sh	Add executable permissions to owner u = owner g = group o = others a = all ```linuxuser@linuxserver:~$ chmod u+x /home/file.txt linuxuser@linuxserver:~$```
sudo chage -m 0 -M 99999 -I -1 -E -1 <username>	-m 0: Sets the minimum days between password changes to 0. -M 99999: Sets the maximum days for password validity to roughly 273 years (effectively never). -I -1: Sets "Password inactive" to never. -E -1: Sets "Account expires" to never. ```linuxuser@linuxserver:~$ sudo chage -m 0 -M 99999 -I -1 -E -1 linuxuser linuxuser@linuxserver:~$ sudo chage -l linuxuser Last password change : Jan 10, 2026 Password expires : never Password inactive : never Account expires : never Minimum number of days between password change : 0 Maximum number of days between password change : 99999 Number of days of warning before password expires : 7 linuxuser@linuxserver:~$```

Sudo nano /etc/security/faillock.conf f	Edit faillock settings (bad idea) 
Dpkg-reconfigure console-setup	Make linux server text larger Prompt 1: enter Prompt 2: enter Prompt 3: enter (or experiment) Prompt 4: terminusBold Prompt 5: 16x32 `linuxuser@linuxserver:~$ sudo dpkg-reconfigure console-setup_`

Passwd -n <days>
 	
<username>
 	
Passwd -n 1 otto.server
	Minimum password age
	
	`linuxuser@linuxserver:~$ sudo passwd -n 1 otto.server`
	`passwd: password changed.`
	`linuxuser@linuxserver:~$ _`

Passwd -x <days>
 	
<username>
 	
Passwd -x 90 otto.server
	Maximum password age
	
	`linuxuser@linuxserver:~$ sudo passwd -x 90 otto.server`
	`passwd: password changed.`
	`linuxuser@linuxserver:~$ _`

Chage -m <days>
 	
<username>
 	
Chage -m 1 bubblefrog
	Minimum password age
	
	`linuxuser@linuxserver:~$ sudo chage -m 1 bubblefrog`
	`linuxuser@linuxserver:~$`

Chage -M <days>
 	
<username>
 	
Chage -M 90 Bubblefrog
	Maximum password age: countdown
	
	`linuxuser@linuxserver:~$ sudo chage -M 90 bubblefrog`
	`linuxuser@linuxserver:~$`

Usermod -L <username>	Lock account
	
	`linuxuser@linuxserver:~$ sudo usermod -L bubblefrog`
	`linuxuser@linuxserver:~$`

Usermod -U
 	
<username>
	Unlock account
	
	`linuxuser@linuxserver:~$ sudo usermod -U bubblefrog`
	`linuxuser@linuxserver:~$ _`

Chage -l <username>	Lists all expiration settings
	
	`linuxuser@linuxserver:~$ sudo chage -l bubblefrog`
	`Last password change : Jan 23, 2026`
	`Password expires : Apr 23, 2026`
	`Password inactive : never`
	`Account expires : never`
	`Minimum number of days between password change : 1`
	`Maximum number of days between password change : 90`
	`Number of days of warning before password expires : 7`
	`linuxuser@linuxserver:~$ _`

<table>
<tr>
<td>Chage -E <date>

<username></td>
<td>Change expiration date


```
linuxuser@linuxserver:~$ sudo chage -E 2026-12-31 bubblefrog
linuxuser@linuxserver:~$ sudo chage -l bubblefrog
Last password change                                    : Jan 23, 2026
Password expires                                        : Apr 23, 2026
Password inactive                                       : never
Account expires                                         : Dec 31, 2026
Minimum number of days between password change          : 1
Maximum number of days between password change          : 90
Number of days of warning before password expires       : 7
linuxuser@linuxserver:~$
```

</td>
</tr>
<tr>
<td>Chage -d

<daysTilForcePwdChg>

Chage -d 0 otto.server

Chage -d 90 otto.server</td>
<td>Max password age (days until forced password changes)

Change pwd at next login

Change pwd in 90 days

Example: bubblefrog must change password at next login.


```
linuxuser@linuxserver:~$ sudo chage -d 0 bubblefrog
linuxuser@linuxserver:~$ _
```

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<tr>
<td>Sudo nano 50-cloud-

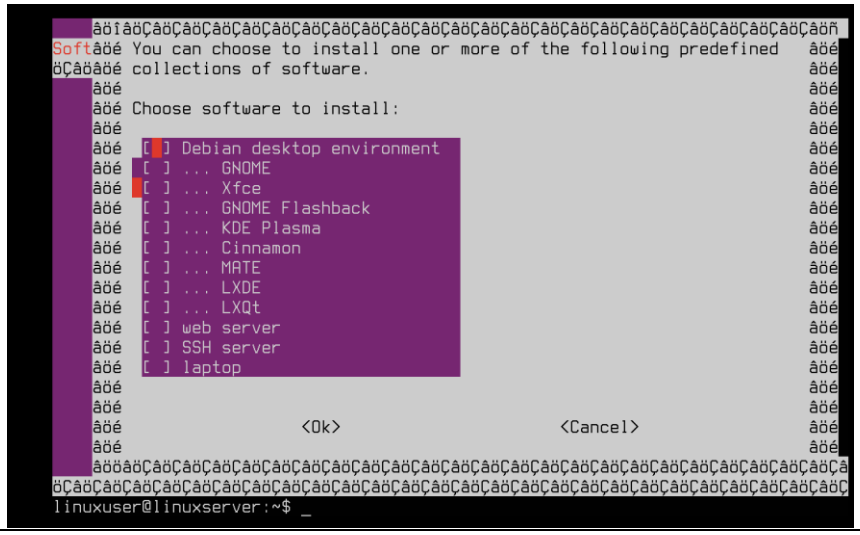
init.yaml</td>
<td>Change IP address on server


```
  GNU nano 7.2                    /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: true
    ens37:
      dhcp4: false
      addresses: [172.10.0.1/16]
      routes:
        - to: default
          via: 192.168.196.2
      nameservers:
        addresses: [172.10.0.2]
        search: [massage.relax]
```

</td>
</tr>
</table>

Sudo netplan apply	Apply IP changes to server
	
	``
	linuxuser@linuxserver:~$ sudo netplan apply
	** (generate:2834): WARNING **: 23:56:47.220: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others
	.
	** (process:2832): WARNING **: 23:56:47.936: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.
	** (process:2832): WARNING **: 23:56:48.365: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.
	linuxuser@linuxserver:~$ _
	``
	
	Switch User
	
	``
	linuxuser@linuxserver:~$ su bubblefrog
	Password:
	You are required to change your password immediately (administrator enforced).
	Changing password for bubblefrog.
	Current password:
	New password:
	Retype new password:
	bubblefrog@linuxserver:/home/linuxuser$ _
	``
	
	Catalog
	
	``
	linuxuser@linuxserver:~$ sudo cat /home/file.txt
	hello
	linuxuser@linuxserver:~$ _
	``
	
	Command used to download data from the web
	
	``
	linuxuser@linuxserver:~$ sudo curl -L http://thinkbroadband.com -o /usr/local/bi
	n/testfile.zip
	  % Total % Received % Xferd Average Speed Time Time Time Current
	                Dload Upload Total Spent Left Speed
	100 162 100 162 0 0 325 0 --:--:-- --:--:-- --:--:-- 325
	100 115k 0 115k 0 0 68617 0 --:--:-- 0:00:01 --:--:-- 393k
	linuxuser@linuxserver:~$
	``
	
	Find
	
	Searches files & directories based on name, size, type,
	
	modification date, owner, and permissions
	
	Standard syntax is find (path) (options) (expression)
	
	OPTIONS
	
	• iname "pattern"
	• type [f | d | l] (file, directory, or link)
	• Size [+/-] [c | M | G] (c = bytes, M = MB, G = GB)

Su	
Sudo cat	
Curl	
Find	

	• Mtime (last modification date) (+/-) (n=days ago)
	• username
	• perm permissions
	• Empty
	• Max depth level
	• and (multiple criteria both must be true)
	• or (either can be true)
	• not (negates the following criterion)
	ACTIONS
	• print
	• ls
	• delete
	• Exec command () \\ (executes a command on each found file [] is a placeholder for the filename. \\; terminates the command.)
	Example
	Sudo find /var/www/html/ -type d -exec
	chmod 750 [ ] \
	`linuxuser@linuxserver:~$ sudo find /usr/local/bin -name "testfile.zip"`
	`/usr/local/bin/testfile.zip`
	`linuxuser@linuxserver:~$ _`

Tasksel	Installs a GUI shell